Group E1

Introduction & Objectives and System Design & Engineering Approach

Md Mahbubur Rahman
Student ID: 1230440

Introduction: Autonomous cars are playing an increasingly vital role day by day in modern engineering, automation and logistics. The main goal of this project is to design, construct and program a two-wheeled autonomous car controlled by an Arduino. The primary objective was to work with Computer-Aided Design (CAD) to print the components and circuit simulation into a functional robotic system that is capable of reacting to its surroundings based on sensor feedback.

Objectives: The final goal of this project was to develop a fully functional robot capable of autonomously navigating a black stripe. The objectives were divided into two main tasks:

- **Task 1- Line Following:** We had to use a stable data collector system using infrared (IR) sensors to smoothly detect the black line on the surface and control motor speeds accordingly to navigate straight paths and turns without being distracted from the line.
- **Task 2- Obstacle Detection and Avoidance:** We used ultrasonic sensors to detect the obstacles in the car's path. So that it can find the open space in its surroundings. And if there is an obstacle, it can execute a maneuver to avoid the obstacle by stop following the line and successfully relocating and following the black line again to continue navigation.
- **Hardware and Structural Design:** We had to design custom 3D-printed mechanical mounts and holders to securely hold sensors, motors and the battery. We tried to ensure optimal weight distribution for the reliable and smooth performance.
- **Software Optimization:** We wrote the control program and simulated it on Tinkercad. The objective was to implement conditional logic to read sensor inputs and trigger the appropriate motor movements for line tracking and obstacle avoidance.

System Design and Engineering Approach: Our team was following a roadmap approach to build a reliable vehicle. This involved sequentially executing steps like initial 3D modeling and virtual simulation, then adjusting all our components and working as a team for the physical prototype. Later multiple tests and debugged.

Hardware Architecture & Mechanical Design: The physical structure was constructed on a wood chassis. Our vehicle was being powered by a battery. The layout of the components was carefully planned to balance weight and improve motor performance.

Sensor Placement: Two ultrasonic sensors were mounted near the front outer edges to provide a wide detection angle. Two IR line sensors were centered between them close to the ground for clear line detection.

Weight Distribution: The heavy battery was positioned centrally between the drive motors to keep the center of mass balanced and avoid overloading the front part and the breadboard on the battery.

Custom Parts (CAD & 3D Printing): Component holders like sensor brackets, motor mounts and the battery frame were modeled in SolidWorks and 3D printed. These parts underwent multiple physical revisions. For example: strengthening fragile ultrasonic mounts and modifying motor brackets to mount the chassis a little bit so the wheels could spin without friction.

Electronics & Circuit Implementation: The electronic system relies on an Arduino board as the central processing unit.

- **Motor Control:** An L293D motor driver was connected between the Arduino and two DC motors.
- **Sensing Unit:** Digital input pins read values from the IR sensors. Ultrasonic sensors are detecting and measure obstacle distances.
- **Power Management:** The breadboard used to power the connections carefully to maintain a steady voltage to the motors and other components.

Prototyping Group E-1 Final Report Part 3-4: Hardware and Software Implementation, Simulation and Testing Results

Muhammad Naimur Rahman Sakib
1233948

July 25, 2026

1 Hardware and Software Implementation

The hardware implementation involved assembling the vehicle using an Arduino, an L293D motor driver, two DC motors, a battery, a breadboard, two IR line sensors, and two ultrasonic sensors. The IR and ultrasonic sensors were mounted at the front for line and obstacle detection, with the ultrasonic sensors positioned near the chassis edges and the IR sensors centered between them. The motors and battery were arranged to maintain balanced weight distribution, while the Arduino and motor driver were placed centrally to reduce wiring complexity. Custom component holders for the motors, sensors, battery, and breadboard were designed in SolidWorks and fabricated using 3D printing. Several designs were refined during assembly, including strengthening the ultrasonic sensor holder, modifying the motor holder to prevent wheel interference, simplifying the IR sensor holder, and redesigning the battery holder to improve accessibility. The Arduino served as the main controller, reading inputs from the IR and ultrasonic sensors and controlling the motors through the L293D driver. Digital pins were used for sensor inputs, while PWM-capable pins controlled motor speed using ‘analogWrite()‘. During testing, a motor speed control issue was resolved by swapping the pins. Software development was performed incrementally by first testing the IR and ultrasonic sensors independently before integrating them into a complete control program developed in Tinkercad. In the final implementation, the vehicle followed a black line using two IR sensors: it moved forward when both sensors detected the white surface and turned when one sensor detected the black line to return to the path.

2 Simulation and Testing Results

We first used Tinkercad to simulate the circuit and check the initial code before testing the real vehicle. The simulation showed that the basic motor and sensor logic worked and helped prepare the wiring diagram for the Arduino, breadboard, motor controller, battery, IR sensors, and ultrasonic sensors. After that, we tested the sensors separately. The IR sensors correctly identified between black and white surfaces, and the ultrasonic sensors produced accurate distance readings in the serial monitor, meaning the pins and the softwere were setup correctly. In Task 1, we focused on basic movement and black-line following. The first hardware test did not work immediately. After checking the circuit several times, we found issues with the breadboard wiring, motor terminal connections, and motor controller wiring around the VIN, 12 V supply, and breadboard. During testing, one Arduino board had a short circuit and had to be replaced. After rewiring and replacing the board, the motors started working. At first, the wheels rotated too quickly, so we moved the motor enable wires to PWM-capable Arduino pins, allowing the motor speed to be controlled. The first line-following result was also unstable. The vehicle detected the black line but initially turned in the wrong direction, so the steering logic was corrected. We also reduced the turning speed to prevent overshooting on curved sections. Some material from the motor holders interfered with the wheels and was removed, while the IR sensors were recalibrated by adjusting their sensitivity potentiometers. By the end of the June and July Task 1 tests, the vehicle could follow the black line smoothly using the two IR sensors and corrected motor speeds. For Task 2, we tested obstacle detection and avoidance. Initially, the ultrasonic sensors only stopped the vehicle when an obstacle was detected within 20 cm. We then improved the code so that the vehicle compared the left and right ultrasonic readings, selected the side with more free space, performed an avoidance maneuver, and searched for the black line again after passing the obstacle. During Task 2 testing, we adjusted the speed values, turning times, and obstacle threshold several times. The system was still sensitive to obstacle position, as sharp turns or closely placed obstacles could cause the vehicle to lose the line temporarily. A later version combined smoother line-following with the improved avoidance algorithm, allowing the vehicle to avoid obstacles, return to the black line, and continue following it. Overall, repeated testing and incremental improvements resulted in successful line detection, obstacle avoidance, and line recovery.

Prototyping Group E-1 Final Report
Part 5-6: Challenges, Conclusions and Future Improvements

Barış Koluaçık
1233167

July 25, 2026

1 Challenges

During the development of our robot, our team encountered numerous obstacles, encompassing both hardware and software issues. These challenges encompassed a wide range from coding errors, malfunctioning electronics, and physical damage due to accidents. As a team, we diligently tackled these issues to ensure that they did not impede the functionality of the robot.

1.1 Hardware Challenges

Our team faced several challenges caused by broken structural parts and defective components. These problems, presented in no particular order, included a Defective motor driver that was causing problems with PWM driving with low duty cycles and had issues with power balancing, thus needed replacement. And whilst testing, an accidental short circuit from a sensor, which caused the Arduino to not power on. Also during development, the 3D-printed part holding the 2 IR sensors fractured when the robot flashed and instantly jumped off the table. A previously printed and unused breadboard holder was repurposed as a sensor holder. Then during the final testing, the robot was accidentally dropped and the left motor mount was broken in a critical way, repair was attempted with additional screws and super glue that did not work, so stock parts supplied by the class were used to replace it before the demonstration.

1.2 Software Challenges

Our team also had to face software related challenges while developing the robot, In our earlier revisions, the ultrasonic sensor was updated constantly, which created overhead issues with the MCU and caused inaccuracies and problems with line tracking. We approached this issue by using millis() to keep time and updating the distance in intervals rather than constantly, thus reducing overhead. Also , we found that controlling the speed of the motors with a constant PWM led to a major problem, the motors weren't being driven correctly. We addressed this issue by integrating a PID loop to adjust the speed of the robot within specified constraints.

2 Conclusions

This project, whilst not performing up to our expectations, gave us firsthand experience in making a closed loop system and dealing with challenges both urgent and non-urgent, diagnosing problems, and we gained experience in embedded development and navigating around the computational limits. This robot was the product of many revisions of the software and many nights and hours of work.

3 Future Improvements

As a team, we always tried to implement features more smartly so that no matter the situation, the robot performs the maneuvers, for the future, we plan to revise the whole structure, use BLDC motors and custom gearing to increase torque on lower speeds, integration of OpenCV or LiDAR based SLAM and a more powerful Raspberry Pi SBC instead of the 8 Bit Arduino with its limited computational capability.